# CS556 Fall 2022 Test 1

Name:

Closed book: no textbook, no electronic devices, one sheet of paper with handwritten notes. *Read carefully before answering!* Write your answers on this test paper. Also hand in your note sheet, with your name on it.

**Question 1** *(10 points)* Let $A = \begin{bmatrix} -2 & 3 \\ 2 & 3 \end{bmatrix}$

(a) Find eigenvalues and an independent set of eigenvectors.

(b) Diagonalize A.

$(A-\lambda I)$ is singular

$\det(A-\lambda I)=0$

$\det\left(\begin{bmatrix} -2 & 3 \\ 2 & 3 \end{bmatrix} - \lambda\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = \det\left(\begin{bmatrix} -2-\lambda & 3 \\ 2 & 3-\lambda \end{bmatrix}\right)$

$(-2-\lambda)(3-\lambda)-6=0$

$\lambda^2-\lambda-12=0$

$(\lambda-4)(\lambda+3)=0$

$\lambda_1=4$

$\lambda_2=-3$

$v \in N(A-4I)$

$\begin{bmatrix} -6 & 3 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow v = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$

$v \in N(A+3I)$

$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow v = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$

**Question 2** *(5 points)* Assume **u** and **v** are unit vectors. What is the value of the dot product $(\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}-\mathbf{v})$?

(a) 2

(b) 1

(c) 0

(d) None of the above, we need to know the vectors **u** and **v**.

$(u+v)\cdot(u-v) = u\cdot u - uv + vu - v\cdot v$

$= 1-1$

$= 0$

**Question 3** *(5 points)* The two vectors $(-1, 1-a^2, 1)$ and $(-1, 1, a^2)$ are orthogonal for all values of $a$:

(a) True

(b) False

$\begin{bmatrix} -1 \\ 1-a^2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 1 \\ a^2 \end{bmatrix} = 2$

**Question 4** *(5 points)* If $A$ and $B$ are $n \times n$ symmetric matrices, that is, $A^T = A$ and $B^T = B$, then $ABA$ is symmetric matrix.

(a) True

(b) False

$(ABA)^T = A^T B^T A^T = ABA$

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**Question 5** *(10 points)* Use Gauss-Jordan elimination to find $A^{-1}$ if

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{R3=R3-R1 \\ R2\cdot(-1) \\ R3\cdot(-1)}]{} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 \end{array}\right] \xrightarrow[R2=R2-R3]{R1=R1-2R3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 0 & 2 \\ 0 & 1 & 0 & -1 & -1 & 1 \\ 0 & 0 & 1 & 1 & 0 & -1 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} -1 & 0 & 2 \\ -1 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix}$$

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**Question 6** *(15 points)* Consider the following system of linear equation for some constants $b$ and $g$:

$$1x + by + 0z = 1$$
$$1x - 2y - 1z = 2$$
$$0x + 1y + 1z = g$$

a) Which number $b$ leads to a row exchange?

b) Which number $b$ makes the system singular (missing a third pivot)?

c) For the $b$ found in part (b), for which value of $g$, the system has infinitely many solutions?

d)Find the solutions of the system with $b$ and $g$ found in parts (b) and (c), if $z = 2$

$$\begin{bmatrix} 1 & b & 0 \\ 1 & -2 & -1 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ g \end{bmatrix}$$

$R2 = R2 - R1$

$$\begin{bmatrix} 1 & b & 0 \\ 0 & -2-b & -1 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ g \end{bmatrix}$$

a) We will have a row exchange if $-2-b=0 \rightarrow b=-2$

$R3 \leftrightarrow R2$

$$\begin{bmatrix} 1 & b & 0 \\ 0 & 1 & 1 \\ 0 & -2-b & -1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ g \\ 1 \end{bmatrix}$$

b) We will miss a pivot for $-2-b=-1 \rightarrow b=-1$

$$\begin{bmatrix} 1 & b & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ g \\ 1 \end{bmatrix}$$

$R3 = R3 + R2$

$$\begin{bmatrix} 1 & b & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ g \\ 1+g \end{bmatrix}$$

c) System with have an infinite number of solutions for $1+g=0 \rightarrow g=-1$

d) If $z=2$, then $x=-2$ and $y=-3$, so the solution is $\begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix}$

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**Question 7** *(10 points)* Construct a matrix with the required property or explain why that is impossible.

(a) Column space contains $\begin{bmatrix}1\\-2\\3\end{bmatrix}$ and $\begin{bmatrix}2\\3\\-5\end{bmatrix}$, nullspace contains $\begin{bmatrix}1\\1\\1\end{bmatrix}$

$$\begin{bmatrix}1 & 2 & v_1\\ -2 & 3 & v_2\\ 3 & -5 & v_3\end{bmatrix}\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

$1+2+v_1=0 \Rightarrow v_1=-3$

$-2+3+v_2=0 \Rightarrow v_2=-1$

$3-5+v_3=0 \Rightarrow v_3=2$

$$\begin{bmatrix}1 & 2 & -3\\ -2 & 3 & -1\\ 3 & -5 & 2\end{bmatrix}$$

**Question 8** *(10 points)* Find the determinant of matrix A:

$$A=\begin{bmatrix}1 & 0 & 1\\ 0 & 3 & 2\\ 1 & -1 & -2\end{bmatrix}$$

$$\det(A)=1^{(1+1)}\cdot 1\begin{vmatrix}3 & 2\\ -1 & -2\end{vmatrix}+1^{1+3}\cdot 1\cdot\begin{vmatrix}0 & 3\\ 1 & -1\end{vmatrix}$$

$$=-4-3$$

$$=-7$$

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**Question 9** *(15 points)* Find a basis for each of the four fundamental subspaces of the matrix.

$$A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

- COLUMN SPACE

$$\begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 5 & 2 & 4 \end{bmatrix} \xrightarrow{R2=R2-2R1} \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix} \xrightarrow{R3=R3-R2} \begin{bmatrix} \textcircled{1} & 3 & 1 & 2 \\ 0 & 0 & \textcircled{2} & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

pivot cols, free cols

basis for $C(A) = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} \right\}$

- NULLSPACE

$$\begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Find the nullspace vectors of the above equation.

basis for $N(A) = \left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$

- Row Space

$$A^T = \begin{bmatrix} \textcircled{1} & 2 & 0 \\ 3 & 6 & 0 \\ 1 & 4 & 2 \\ 2 & 8 & 4 \end{bmatrix} \xrightarrow[R3=R3-R1]{R2=R2-3R1} \xrightarrow{R4=R4-2R1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 4 & 4 \end{bmatrix} \xrightarrow{R2 \leftrightarrow R4} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 4 & 4 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R3=R3-2R2} \begin{bmatrix} \textcircled{1} & 2 & 0 \\ 0 & \textcircled{4} & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

basis for $C(A^T) = \left\{ \begin{bmatrix} 1 \\ 3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ 4 \\ 8 \end{bmatrix} \right\}$

- Left Null Space

$$\begin{bmatrix} \textcircled{1} & 2 & 0 \\ 0 & \textcircled{4} & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$c = 1, b = -1, a = 2$

basis for $N(A^T) = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

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**Question 10** *(15 points)* Let $a_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ and $a_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$.

(a) Find the projection of the vector $b = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$ onto the subspace spanned by $a_1$ and $a_2$.

(b) Find the error vector and its magnitude.

(c) Use Gram-Schmidt process to find an orthonormal basis for the span of $a_1$ and $a_2$.

a) $p = A(A^TA)^{-1}A^Tb$

$$A^TA = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$$

$$(A^TA)^{-1} = \frac{1}{6}\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}$$

$$p = \frac{1}{6}\begin{bmatrix} 1 & 1 \\ -1 & 0 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}\begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2/6 \\ 7/6 \\ 11/6 \end{bmatrix}$$

$$= \frac{1}{6}\begin{bmatrix} 2 & 0 \\ -5 & 3 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 4 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 2 \\ 7 \\ 11 \end{bmatrix}$$

$$= \begin{bmatrix} 2/6 \\ 7/6 \\ 11/6 \end{bmatrix}$$

b) $e = b - p = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 2/6 \\ 7/6 \\ 11/6 \end{bmatrix} = \begin{bmatrix} -2/6 \\ -1/6 \\ 1/6 \end{bmatrix}$

c) $x = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

$$y = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - \frac{x^Ty}{x^Tx}x = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - \frac{3}{3}\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$q_1 = \frac{x}{\|x\|} = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$q_2 = \frac{y}{\|y\|} = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

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